



Department of Computer Science and Business Systems

Academic Year 2023– 2024 (Odd/ Even Semester)

Degree, Semester & Branch: B.Tech, IV & CSBS

Course Code & Title: AL3452 & Operating Systems

Name of the Faculty member(s): Dr.M.GomathyNayagam

Innovative Practice Description

- **Unit/Topic:**Unit I/ System Calls
- **Course Outcome:** CO1
- **Topic Learning Outcome:** UO1b
- **Activity Chosen:** Flash Card
- **Justification:**

Because Flashcard active learning is so effective at summarizing material, encouraging repetition and recall, improving learning efficiency, providing flexibility, raising engagement, enabling self-paced learning, and acting as a diagnostic tool, its use in teaching System Calls in Operating Systems is justified. The combination of these elements results in a thorough and efficient learning process for students learning about system calls in operating systems.

- **Time Allotted for the Activity:**30 Minutes
- **Details of the Implementation:**
 - Questions and Answer cards were prepared in chart paper by the course instructor which is shown in fig.3.
 - The entire question cards were given to one particular student (Mr.Balaji) shown in fig.1.
 - The entire class divided into 3 groups.
 - Each group could have 3 answer cards.
 - The answer cards were distributed to the group by the course instructor.
 - Questioner asked the questions one by one in the class.
 - Questioner gave 2 minutes time to each question for identifying the answer card to the group.

- The members of the group heard the question and found out the corresponding answer card which they were having based on the topic discussed in class on the day.
- Any one of the group found the answer card with them, one of the students from the group came to the stage and he/she explained the answer for the question within 2 minutes is shown in figure 2.

• **CO–PO/ PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO 1	3	3	3					1	1	1		1	1	1	

(1– Low 2–Moderate 3 – High)

• **PO/ PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO8	PO9
	3	3	3	1	1
Justification for correlation	The students could be able to apply the basic engineering knowledge to use the various system calls	The students could be able use the system calls to solve the various issues in process, device, memory, file management and etc.	The students could able to provide the solution for identifying the relationship between system calls and various OS services	The students could be able to answer the question during the event in ethical way.	The students can be able to discuss with their peer members and answer for the particular question during the flash card activity.

Innovative practice	PO10	PO12	PSO1	PSO2
	1	1	1	1
Justification for correlation	The students can be able to discuss with their peer members and answer for the particular question	The students could be able to continuously learn about the systems calls.	The students could be able to apply the various systems calls during OS usage and iOS for the various	The students could be able to apply the various systems calls during OS usage and iOS for the various

	during the flash card activity.		services of OS	services of OS
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- Images/Screen shot of the practice:



Fig.1. Mr.Balaji (Questionnaire) ask the Questions to the Class

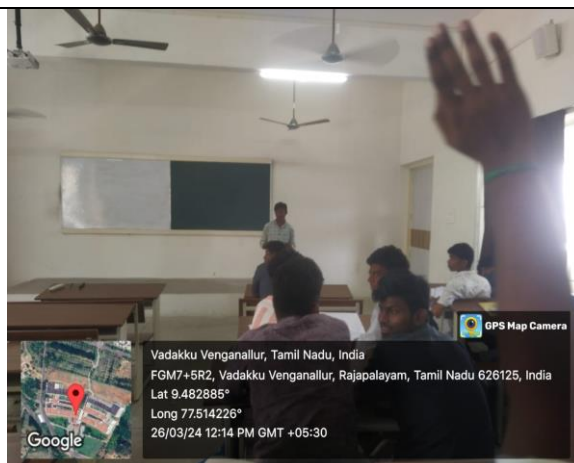


Fig.2. Mr.Sakthivel from Group 3 Answered the question by raising his hand

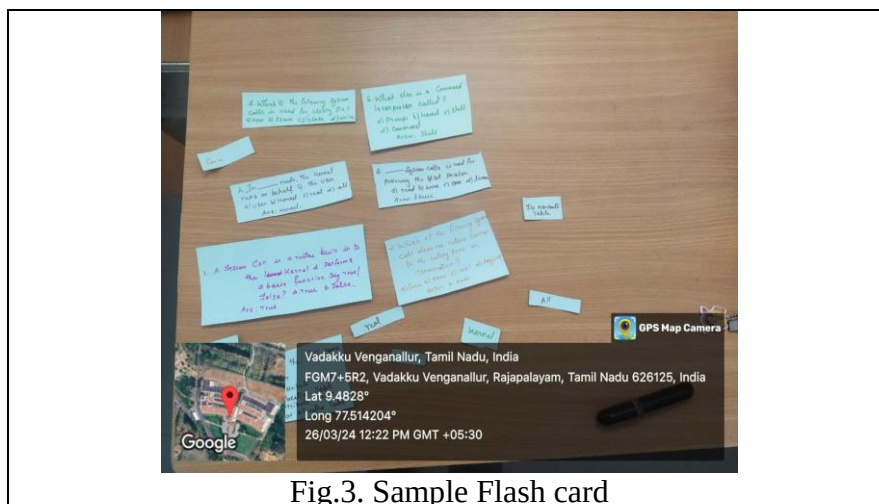


Fig.3. Sample Flash card

- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

All the students are involved to answer for the question asked by the questionnaire.

- ❖ **Benefit of the Practice (E.g. Outcome Attainment would have increased due to innovative practice over conventional practice)**

- It helps students to know and recap all the system calls usage

- ❖ **Challenges faced in implementation:**

- Make all students to participate for answering the questions.

References:

- ❖ Mr.M.GomathyNayagam, “Functional Requirements of OGSA/OGSI” in course CS6703 Grid and Cloud Computing, https://www.ritrjpm.ac.in/images/computer-science/13_CS6703_Falshcards.pdf
- ❖ <https://www.teachingenglish.org.uk/article/using-flash-cards-young-learners>
- ❖ <https://voxy.com/blog/2011/05/are-flashcards-an-effective-learning-tool-infographic/>
- ❖ <https://elearningindustry.com/the-5-best-free-flashcard-creation-tools-for-teachers>

Signature of Faculty Member
(Dr.M.Gomathy Nayagam)

HOD